SYSTEM REQUIREMENT SPECIFICATION

CraftConnect

Requirement Analysis & Product Planning Document

Milestone 1
Timeline: 30 Jul – 14 Aug
Status: Draft / Approved

1. PROBLEM STATEMENT

Students and beginner freelancers with creative or technical skills (art, design, video editing, writing, code, music) struggle to find a low-barrier way to earn income from their work. General freelance platforms are saturated with professionals, buried in heavy commission fees, or increasingly flooded with AI-generated submissions that undercut genuine human skill.

CraftConnect solves this by giving students a dedicated marketplace where **only human-created work** is offered, making it easier for beginners to get discovered and for clients to trust what they are paying for.

2. REFINED OBJECTIVES

- **Creator Profiles & Services:** Allow students/freelancers to build portfolios, list skills, set price tiers, and accept commission requests.
- **Human-Centric Guarantee:** Enforce and signal "100% human-made" work as the platform's core differentiator and trust model.
- **Discovery & Marketplace:** Enable clients to discover talent via category filtering, compare portfolios and pricing, and request custom work.
- **Order Lifecycle Pipeline:** Provide a structured workflow: `Request` → `Accept/Reject` → `In-Progress` → `Delivered` → `Reviewed`.
- **Trust & Reputation System:** Build credibility through a ratings and reviews system tied strictly to verified, completed orders.

3. USER ROLES (ACTORS)

Role	Description & Permissions
Guest	Unauthenticated visitor. Can browse creator profiles, explore categories, and search listings; cannot place commission requests.
Customer	Registered client user. Can request custom commissions, submit reference files, make payments, track status, and leave reviews.
Creator	Registered talent user. Builds public portfolio, accepts/rejects commission requests, delivers files, and manages earnings.
Admin *(Stretch)*	Platform moderator. Reviews reported profiles, moderates content listings, and resolves commission disputes.

Note: A single user account holds both Customer and Creator capabilities, avoiding rigid account switching.

4. FUNCTIONAL REQUIREMENTS

- **FR1 (Auth):** User registration/login via email+password and Google OAuth (Supabase Auth).
- **FR2 (Profile):** Creator profile editing: bio, skill tags, price ranges, and portfolio media uploads.
- **FR3 (Discovery):** Browse and search creators by skill category, keywords, and pricing filters.
- **FR4 (Public View):** Showcase public creator profile
- **FR6 (Request Handling):** Creator workflow to accept or decline incoming customer briefs.
- **FR7 (Order Pipeline):** Real-time status tracking: `Pending` → `Accepted` → `In Progress` → `Delivered` → `Completed`.
- **FR8 (Ratings & Reviews):** Post-completion 1–5 star rating + written review validation.

5. NON-FUNCTIONAL REQUIREMENTS

Attribute	Specification & Quality Standards
Security	Hashed passwords via Supabase Auth. Enforce Row-Level Security (RLS) policies on Postgres tables to ensure users can only modify their owned records.
Usability	Fully mobile-responsive layout. Navigation rule: Max 3 clicks from homepage landing to any creator's profile page.
Performance	Portfolio image assets compressed and lazy-loaded. Database search queries executed in < 1s for standard data sets.
Data Integrity	Strict relational constraints: A review record can only be instantiated if linked to a valid commission marked as `Completed`.
Maintainability	Utilize managed BaaS (Backend-as-a-Service) to eliminate infrastructure overhead for a small team, ensuring delivery within timeline.

6. USE CASES SUMMARY

ID	Use Case Name	Primary Actor	Trigger / Description
UC1	Register / Authenticate	Guest	User accesses system to browse fully or initiate transactions.
UC2	Build Profile & Portfolio	Creator	Creator sets up skills, bio, pricing, and sample gallery.
UC3	Search & Filter Talent	Customer / Guest	User searches for specific creative skills or price points.
UC4	View Creator Profile	Customer / Guest	User clicks a creator card to inspect portfolio and ratings.
UC5	Submit Commission Request	Customer	Customer sends project brief, budget, and deadline.
UC6	Manage Order Request	Creator	Creator reviews brief and chooses to accept or reject.
UC7	Track Order Progress	Customer / Creator	Actors monitor order lifecycle through status stages.
UC8	Leave Rating & Review	Customer	Order reaches `Completed` status; client leaves feedback.
UC9	View Role Dashboard	Customer / Creator	User accesses overview of active work, history, and earnings.

7. TECHNOLOGY STACK EVALUATION

OPTION A: Fast Track — RECOMMENDED

- **Frontend:** React (Vite) + TailwindCSS
- **Backend / DB:** Supabase (Postgres, Auth, Object Storage, RLS)
- **Hosting:** Vercel (Frontend Deployment)

Rationale: Cuts backend boilerplate drastically. Relational Postgres fits commissions and reviews perfectly. Scaffolds natively with modern AI tools, drastically increasing build speed across Milestones 2–4.

OPTION B: Classic MERN — ALTERNATIVE

- **Frontend:** React.js
- **Backend:** Node.js + Express REST API
- **Database:** MongoDB (NoSQL)
- **Hosting:** Render / Railway + Vercel

Rationale: High resume value for manual API development, but consumes significantly more setup time. Introduces higher timeline risk against the submission deadline.

8. DATABASE SCHEMA & ENTITY RELATIONSHIPS

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users ├─ id (PK, uuid) ├─ name, email, password_hash (Managed by Supabase Auth) ├─ role_flags (is_creator, is_customer) └─ created_at (timestamp) creator_profiles ├─ id (PK, uuid) ├─ user_id (FK → users.id) ├─ bio, category_id (FK → categories.id) ├─ price_range, skills (array) ├─ avg_rating (numeric, cached) └─ created_at (timestamp) categories ├─ id (PK, int) └─ name (Digital Art, Graphic Design, Video, Writing, Code, Music) portfolio_items ├─ id (PK, uuid) ├─ profile_id (FK → creator_profiles.id) ├─ title, description, image_url └─ created_at (timestamp) commissions ├─ id (PK, uuid) ├─ customer_id (FK → users.id), creator_id (FK → users.id) ├─ category_id (FK → categories.id) ├─ description, budget, deadline ├─ status (pending | accepted | rejected | in_progress | delivered | completed | cancelled) └─ created_at, updated_at (timestamp) reviews ├─ id (PK, uuid) ├─ commission_id (FK → commissions.id, UNIQUE 1:1) ├─ reviewer_id (FK → users.id) ├─ rating (int 1-5), comment (text) └─ created_at (timestamp)
```

Key Relationships: `users 1—1 creator_profiles` | `creator_profiles 1—N portfolio_items` | `users 1—N commissions` | `commissions 1—1 reviews`

9. WIREFRAME & LAYOUT ARCHITECTURE

The system layout architecture is documented in `CraftConnect_Wireframes`. The low-fidelity wireframe set covers structural layouts for core application screens to lock UX decisions prior to Milestone 2 development:

- **Home / Marketplace:** Hero section with search bar, category pills, and creator card grids.
- **Creator Profile Page:** Header banner, bio, skill tags, portfolio grid, and direct commission request trigger button.
- **User Dashboard:** Split view for active requests (as creator) and active orders (as customer) with status trackers.
- **Auth Flow:** Clean modal/page forms for single-account login/registration.

10. MILESTONE 1 CHECKLIST & ACTION PLAN

- ✓ Problem statement & refined objectives finalized
- ✓ Functional + non-functional requirements documented
- ✓ Use cases & actor responsibilities mapped
- ✓ Tech stack finalized (Recommended: Supabase + React + TailwindCSS)
- ✓ Relational Database Schema (ERD) drafted
- ✓ Low-fidelity structural wireframes completed
- ○ **Next Step (Milestone 2: 15 Aug – 29 Aug):** Initialize project scaffolding, set up Supabase schema/RLS, and build core frontend pages (Home, Login, Profile, Dashboard).